AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Final Examination
Spring Semester 2025/2026

Date: 8 /6 /2026

AOU EGYPT

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: 2:00 Hours**

Instructions:

- Total Marks: **50 marks**
- This exam consists of 5 questions
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

You should answer all the questions, giving 50 possible marks. Show your work.

Q1. [2+2+3+ 3 Marks]

a) Given the function f(x) = 6+ 6x -3 x^2

i) Find the domain of the function f.

ii) Find the axis of symmetry and find the maximum or minimum value of the function f.

iii) Find the zeros for the quadratic function f(x).

b) Use synthetic division $to\ devide\ f(x) = 3x^6 - 2x^5 - x^4 - x^2 - 2x + 3$ by x -1 and write the quotient and the remainder.

Q2. [3+4+3 marks]

a) Find the equation of the line passes through the points (0, 0) and (-3, 1)

b) Given the function f(x) =x^2 +1, determine if the f is one to one function.

c) If z varies jointly as x and y, and z= 18 when x= 0.4 and y=3 find z when x= 1.2 and y=2.

Q3. [4+2+4marks]

a) Solve the inequality: $x^3 - 9x > 0$

b) Find the domain of the following and write the answer in interval notation

i) F(x)= ln lxl

ii) G(x)= $\sqrt{3x + 21}$

Q4: [3+3+4 marks]

a) Solve for x:

i) $2\ln x = \ln(x + 1)$

$ii)\ \log_4(5x - 13) - \log_4(x - 2) = 1$

b) If f(x)= $\log_3(x)$ and g(x)= 9^x, find f (g (3)).

Q5. [2+2+3+3 marks]

a) How many 3-digit numbers can be formed using the digits 1, 3, 5, and 9 if repetitions are allowed.

b) The yes, No and undecided responses to a survey question are broken down according to employment status and the sample results are given below.

Employment Status	Yes	No	undecided	Total
Employed	18	53	4	75
Unemployed	12	7	6	25
Total	30	60	10	100

If a person is picked at random from this group, find the probability that:

i) The person is Employed and he says yes.

ii) The person is unemployed or he says yes.

iii) The person is employed , given that he says yes.

End of question

MOE EGYPT